LETTER TO THE EDITOR

INTEGRATIVE ZOOLOGY

Chromosome-level genome of *Scolopendra mutilans* provides insights into its evolution

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Arthropoda, a diverse phylum encompassing Myriapoda, Crustacea, Chelicerata, and Insecta, constitutes more than 80% of the total number of documented species (Qu *et al.* 2020). Within these groups, centipedes (Chilopoda) hold a unique position as one of the oldest terrestrial venomous groups, with a fossil history spanning 430 million years. This lineage encompasses five orders and includes more than 3500 species (Undheim & King 2011). Centipedes, characterized by having a pair of legs per segment, have drawn significant attention in the biomedical field (Undheim *et al.* 2015) due to their production of venoms in specialized venom glands that have modified the first pair of trunk legs (Giribet & Edgecombe 2019). Centipedes are carnivorous, soil-dwelling invertebrates with a predatory nature, playing a pivotal role as indicators of soil biodiversity and serving as vital tools for evaluating ecosystem health (Dugon 2017; Halpin *et al.* 2021). Despite their ecological value, the evolutionary ecology of myriapods has received relatively less attention compared to other arthropods. However, recent advances have been made, with nonchromosomal-level genomes of five millipedes and four centipedes having been published, shedding more light on their evolutionary history (Chipman *et al.* 2014; Kenny *et al.* 2015; Qu *et al.* 2020; So *et al.* 2022).

The genus *Scolopendra*, comprising a diverse range of scolopendromorph centipedes, inhabits various regions spanning from tropical to warmer temperate areas worldwide (Lewis 1981). This genus comprises more than 90 documented species (Bonato *et al.* 2016), exhibiting a wide range of coloration, segment variations, and sizes. *Scolopendra* species primarily prey on insects and other invertebrates, playing a crucial role in soil ecosystems (Siriwut *et al.* 2016). Notably, female *Scolopendra* exhibit parental care, an intriguing behavior within the genus (Cabanillas *et al.* 2019). The venom of *Scolopendra* is rich in peptides, contributing to their effectiveness as predators and their ecological success (Yoo *et al.* 2014; Hakim *et al.* 2015).

Through peptidomics and cDNA library analysis, 79 distinct peptide toxins were identified in *Scolopendra subspinipes mutilans* (Rong *et al.* 2015), thereby contributing to the enrichment of centipede venom diversity (Yang *et al.* 2012). Among these toxins, 29 were identified as neurotoxins that target Na^+, K^+, and Ca^{2+} channels

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(Yang *et al.* 2012; Rong *et al.* 2015). These peptide toxins were systematically classified into 17 families based on their sequences and cysteine arrangement (Luo *et al.* 2022). In a comparative proteomic analysis of different body tissues, a total of 1639 proteins were identified in *S. s. mutilans* employing proteotranscriptomics. This included 1204 unique proteins in the torso tissue and 165 unique proteins in the venom gland, with 100 unique proteins overlapping between the two tissues (Zhao *et al.* 2018). However, due to the lack of a genome for this species, studies on the biosynthesis of its venom have been limited.

Prior to assembling the genome of *Scolopendra mutilans* using sequencing data, estimates of its characteristics were conducted. The genome size of *S. mutilans* was estimated to be approximately 1036.6 Mb, based on 196.2 Gb of sequencing data at 189.27 fold coverage. Heterozygosity was 0.57% and GC content at 30.86%, using *k-mer* ($k = 19$) methods (Fig. S1 and Table S1-1, Supporting Information). Additionally, flow cytometry estimation showed the genome size of *S. mutilans* from 1018.75 to 1023.57 Mb (Fig. S2 and Table S1-2, Supporting Information). The subsequent genome assembly produced a genome of 903.91 Mb, containing 352 contigs with a high contig N50 of 24.13 Mb and longest contig of 69.05 Mb, derived from 163.80 Gb of clean bases at 181.21 fold coverage (Tables S1-4,S2 and Fig. S3, Supporting Information).

Notably, the contig N50 achieved in this assembly surpassed that of other myriapods (Chipman *et al.* 2014; Kenny *et al.* 2015; Qu *et al.* 2020; So *et al.* 2022). By combining data from the Illumina, ONT, and Hi-C technologies, we obtained a total of 170.26 Gb of clean data with a 188.36-fold coverage (Tables S1-3,S1-4, Supporting Information). This resulted in 903.93 Mb of genome sequences anchored into 14 chromosomes, with an N50 of 62.8 Mb (Fig. 1a,b). This outcome aligns with previous chromosome karyotype studies, which reported a chromosome count of $2n = 28$ (Ogawa 1953). The anchored sequences accounted for 95.64% of the total contig length (Fig. 1a; Table S1-4, Supporting Information). A heatmap generated from Hi-C data supported the division of all data bins into 14 chromosomes (Fig. 1b).

The estimated genome size of *S. mutilans* falls within the range typical for centipedes, which ranges from 176.2 Mb to 3.2 Gb (So *et al.* 2022). According to the Benchmarking Universal Single-Copy Orthologs (BUSCO) assessment, an impressive 97.0% of genes were identified as complete, comprising 96.3% single-copies genes and 0.7% duplicated genes, while 2.1% were classified as fragmented genes in *S. mutilans*. Remarkably, this achievement represents the highest value among all sequenced genomes of Myriapoda species (Table S2, Supporting Information) (So *et al.* 2022). The combination of a lengthy contig N50 and the high BUSCO score suggests that the current assembly represents a high-quality genome for *S. mutilans*. Centromere location and length were accurately predicted (Fig. S4a and Table S3, Supporting Information). The analysis revealed the presence of nine acrocentric chromosomes and four submetacentric chromosomes (Table S3, Supporting Information), consistent with the previous findings (Ogawa 1953). Remarkably, these results corroborate the earlier karyotype study, indicating that the chromosomes in *S. mutilans* exhibit a straightforward rod-shaped structure.

Based on a combined strategy using *ab initio*, homology-based, and RNA sequence-based prediction, we successfully identified 20 153 protein-coding genes in *S. mutilans* (Table S4, Supporting Information). These genes exhibited an average gene length of 18423.87 bp, with an average of 7.89 exons per gene (Tables S4-1,S5, Supporting Information). Among the 20 153 predicted genes, 17 823 genes (88.44%) were annotated in the Kyoto Encyclopedia of Genes and Genomes (KEGG, 23.95%), KEGG pathway (17.60%), NR (73.91%), Uniport (75.37%), gene ontology (GO, 56.88%), Pfam (61.40%), and Interproscan (84.07%) databases, respectively (Table S6 and Fig. S4b, Supporting Information).

As previously reported, transposable elements (TEs) significantly contribute to the genome size of arthropods (So *et al.* 2022). In the case of *S. mutilans*, 424.98 Mb, constituting 47.01% of the genome, consists of repeat sequences (Table S7-1, Supporting Information). The most abundant categories of TEs include DNA transposons (14.25%), long interspersed nuclear elements (5.38%), and long terminal repeats (5.36%) (Table S7-1 and Fig. S4c, Supporting Information). It is noteworthy that for myriapods, the length of TEs exhibited a positive correlation with genome size, except for short interspersed elements (SINEs) (Fig. S3, Supporting Information). In addition to protein-coding genes, non-coding genes were also identified, which include 83 microRNAs (miRNAs), 955 rRNAs, 5054 tRNAs, and 1207 snRNAs (Table S7-2, Supporting Information).

A comparative genomic analysis was carried out using genomic data from 13 sequenced arthropod species and *S. mutilans*. With *Naja naja* as an outgroup, a genome-wide phylogenetic analysis was conducted on 293 single-copy genes from 14 arthropods (Fig. 1c). The results revealed that *S. mutilans* forms a sister group with *Rhysida immarginata* within Scolopendridae (Scolopendromorpha) and their divergence occurred at approximately 204 Mya. Surprisingly, *S. maritima* (Geophilomorpha) was found to be a sister group with *Lithobius niger* (Lithobiomorpha),

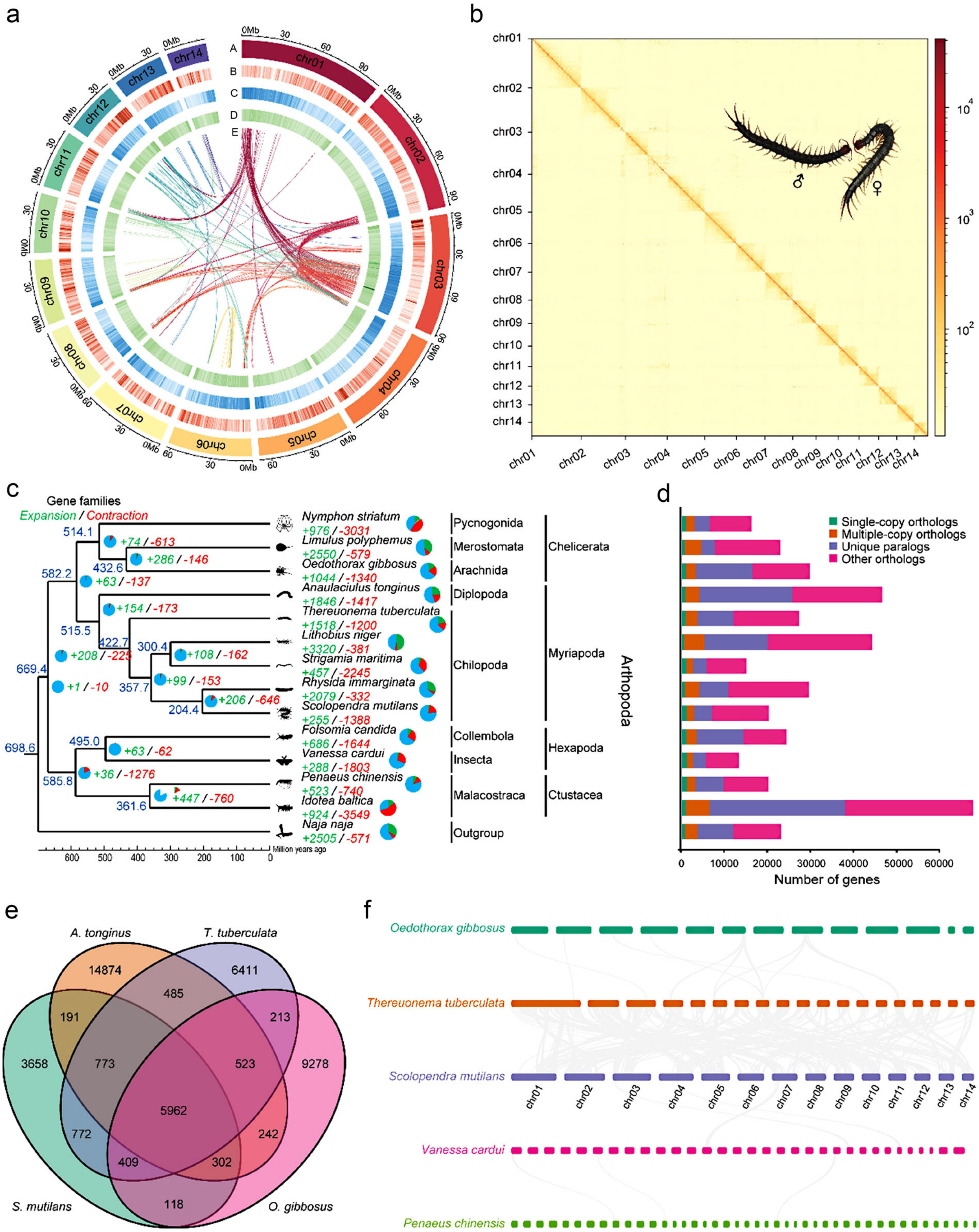

Figure 1 Genome further. (a) Circos graph (from outside to inside) representing chromosome number (A), the gene density (B), all repeat sequence density (C), the GC content distribution (D), and synteny relationship (E) across the chromosomes of the genome with

with their divergence occurring around 300 Mya. This topological structure is consistent with a previous study based on the mitochondrial genomes, which indicated that Lithobiomorpha and Geophilomorpha are sister groups (Hu *et al.* 2020; Ding *et al*. 2022; Yang *et al.* 2023). This finding contrasts with the traditional view that Geophilomorpha and Scolopendromorpha are sister groups based on morphological similarities (Fernández *et al.* 2016). Within Chilopoda, a divergence between the Notostigmophora clade and the Pleurostigmophora clade occurred approximately 423 Mya, closely aligning with the fossil record (Undheim & King 2011).

Gene family clustering analysis of the genome of *S. mutilans* revealed 1177 single-copy genes, 1727 multiple-copy genes, 4139 unique genes, and 13 110 other genes (Fig. 1d). *S. mutilans* shares 7916 (64.97%) gene families with *Thereuonema tuberculata*, 7228 (59.32%) gene families with *Anaulaciulus tonginus*, and 6791 (55.73%) gene families with *Oedothorax gibbosus*, respectively (Fig. 1e). Moreover, 5962 gene families are common to all four arthropods (Fig. 1e). Interestingly, the number of shared gene families exhibits a positive correlation with their phylogenetic relationships.

The genome synteny analysis reveals a high level of synteny relationship between *T. tuberculata* and *S. mutilans* (Fig. 1f), supporting previous work that demonstrates significant genome collinearity among myriapod (So *et al.* 2022).

The gene family evolutionary analysis unveiled that 255 gene families (comprising 1466 genes) had expanded, accounting for only 2.09% (255 out of 12 185) of all families, while 1388 gene families (1553 genes) had contracted in the *S. mutilans* genome compared to its most recent common ancestor (Fig. 1c; Table S4, Supporting Information). The expanded gene families of *S. mutilans* exhibited significant enrichments in 23 GO terms and 14 KEGG pathways (Tables S8,S9, Supporting Information). For the unique gene families of *S. mutilans*, these enrichments predominantly involved catalytic activity (including GTPase activity, hydrolase activity, and O-acyltransferase activity), transporter activity (as channel activity, hexosyltransferase activity, and transmembrane transporter activity), binding (consisting ATP binding, DNA binding, metal ion binding, and protein dimerization activity) (Fig. S5a and Table S10, Supporting Information), and Biosynthesis of unsaturated fatty acids, Fatty acid elongation, Fatty acid metabolism, and Dorso-ventral axis formation (Table S11, Supporting Information). Additionally, these expanded gene families displayed enrichments in pathways such as Steroid biosynthesis, Toll and Imd signaling pathway, Taurine and hypotaurine metabolism, Ubiquitin medicated proteolysis, and Dorso-ventral axis formation (Fig. S5b and Table S8, Supporting Information). In terms of biological processes, genes from significantly expanded gene families were enriched in the metabolism process, such as lipid catabolic process, glutathione catabolic process, and DNA integration (Fig. S5b and Table S9, Supporting Information). Considering the enrichment of orthologs in several of these pathways related to lipid metabolism, immunity, and signal transduction, it is plausible that these genetic adaptations may have contributed to sensory and locomotory adaptations, facilitating the ecological shift to predation in *S. mutilans*, similar to other centipedes (So *et al.* 2022). The results are similar to the findings in *Mesobuthus martensii* (Cao *et al.* 2013) and *Whitmania pigra* (Tong *et al.* 2022).

In the branch model, *S. mutilans* was assigned as the foreground branch, with other myriapods serving as the background branches. A significant difference was observed in the evolution rate of 38 genes between *S. mutilans* and other myriapods. These genes demonstrated significant enrichments in four GO terms (Table S12, Supporting Information), specifically carbohydrate metabolic process, nucleotide-excision repair, regulation of transcription, DNA-templated, and cell periphery.

In animals, the ancestral Hox homologs have been confirmed in arthropods, including labial (*lab*), proboscipedia (*pb*), deformed (*Dfd*), sex combs reduced (*Scr*), fushi tarazu (*ftz*), antennapedia (*Antp*), ultrabithorax (*Ubx*), abdominal-A (*Abd-A*), abdominal-B (*Abd-B*), and zerknüllt (*zen*) (Akam 1998). Within *S. mutilans*, we identified nine Hox cluster proteins, including the antennapedia complex (*lab*, *pb*, *Dfd*, *Scr*, ftz, and *Antp*) (Calhoun *et al.* 2002) and the bithorax complex (*Ubx*, *Abd-A*, and *Abd-B*) (Maeda & Karch 2006). Notably, all the Hox cluster genes are exclusively present on a single chromosome and exhibit a conserved order in centipedes (Figs S6,S7 and Table S13, Supporting Information). These

1-Mb sliding window size. (b) Interaction matrix across the *Scolopendra mutilans* genome; blocks with higher color intensity indicate stronger contacts. (c) Phylogenetic tree and orthology between genomes of *S. mutilans* and 12 other arthropods. The phylogenetic tree was constructed based on 293 single-copy gene families using PhyML. The green numbers indicate expansions, and the red numbers denote contractions. The blue numbers indicate the diversitree time. (d) Different types of orthology with different colors. (e) Venn diagram of the orthologous gene families from four arthropods. (f) Genome synteny among *Oedothorax gibbosus*, *Penaeus chinensis*, *S. mutilans*, *Thereuonema tuberculata*, and *Vanessa cardui*.

structures play a key role in maintaining the relationship between phenotype and genomes (Holland 2013; So *et al.* 2022). Similar to *L. niger*, *R. immarginata*, and *T. tuberculata*, *S. mutilans* lacks *Hox3*, but the *Hox3* family has dynamic changes in different myriapod lineages (So *et al.* 2022).

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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SUPPLEMENTARY MATERIALS

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1-1 The result of k-mer analysis

Table S1-2 Estimated genome size by flow cytometry

Table S1-3 Sequencing and quality filtering statistics

Table S1-4 Scolopendra subspinipes mutilans genome assembly statistics and assembly completeness

Table S2 Comparison of myriapod genome assembly quality

Table S3 The statistical information of mapping using Hi-C data and centromere position of *S. mutilans*

Table S4-1 Basic statistical results of gene prediction

Table S4-2 Statistics results of gene fiamly cluster

Table S5 Statistics results of coding gene

Table S6 Statistics of gene anotation using different database

Table S7-1 Statistical results for transposable elements

Table S7-2 Non-coding RNA content of the *S. mutilans* genome

Table S8 GO enrichments based on expanded gene families of *S. mutilans*

Table S9 KEGG enrichments based on expanse gene families of *S. mutilans*

Table S10 GO enrichments based on unique gene families of *S. mutilans*

Table S11 KEGG enrichments based on positive selected genes of *S. mutilans*

Table S12 GO enrichments based on positive selected genes of *S. mutilans*

Table S13 Hox cluster genes characterized from the *S. mutilans*

Figure S1 GenomeScope k-mer profile plot for the genome of S. mutilans, based on 19-mers in Illumina reads.

Figure S2 Flow cytometry analysis for DNA contents of *S. mutilans*.

Figure S3 The relationship between genome size and TE length in myriapod.

Figure S4 Genome phylogeny and genome evolution.

Figure S5 Enrichment of unique (A) and expanded (B) gene families.

Figure S6 Synteny relationship.

Figure S7 The phylogenetic trees for the Hox cluster.

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